

In-Class Test – 01

Student Number – 35040

Q1

```
public class NetworkDevice {
    String deviceID;
    String ipAddress;
    String deviceName;
    boolean isOnline;
    int threatLevel;

    NetworkDevice(String id, String ip, String name, boolean online, int threat) {
        deviceID = id;
        ipAddress = ip;
        deviceName = name;
        isOnline = online;
        threatLevel = threat;
    }

    void displayStatus() {
        System.out.println("Device ID: " + deviceID);
        System.out.println("Device IP Address: " + ipAddress);
        System.out.println("Device Name: " + deviceName);
        System.out.println("Online Status: " + isOnline);
        System.out.println("Threat Level: " + threatLevel);
    }

    boolean isCritical() {
        return threatLevel >= 7;
    }
}

Run | Debug
public static void main(String[] args) {
    NetworkDevice device1 = new NetworkDevice(id: "D01" , ip: "192.168.1.09" , name: "Router" , online: true , threat: 7);
    device1.displayStatus();
    System.out.println("Is Critical: " + device1.isCritical());
}
}
```

```
ng\Code\User\workspaceStorage\7e18229a918905c06a
Device ID: D01
Device IP Address: 192.168.1.09
Device Name: Router
Online Status: true
Threat Level: 7
Is Critical: true
PS D:\NSBM\1 Year\2 Sem\Java\In class\1>
```

Q2

```
1 public class UserAccount {
2     private String username;
3     private String emailAddress;
4     private String accountStatus;
5     private int failedLoginAttempts;
6     private String password;
7
8     public UserAccount(String username, String emailAddress, String accountStatus, int failedLoginAttempts, String password) {
9         this.username = username;
10        this.emailAddress = emailAddress;
11        this.accountStatus = accountStatus;
12        this.failedLoginAttempts = failedLoginAttempts;
13        this.password = password;
14    }
15
16    public String getUsername() {
17        return username;
18    }
19
20    public void setUsername(String username) {
21        this.username = username;
22    }
23
24    public int getFailedLoginAttempts() {
25        return failedLoginAttempts;
26    }
27
28    public void setFailedLoginAttempts(int failedLoginAttempts) {
29        if (failedLoginAttempts < 0) {
30            System.out.println("Invalid value: Failed login attempts cannot be negative.");
31        } else {
32            this.failedLoginAttempts = failedLoginAttempts;
33        }
34    }
35
36    public void login(String enteredPassword) {
37        if (enteredPassword.equals(password)) {
38            System.out.println("Login successful.");
39            failedLoginAttempts = 0;
40        } else {
41            failedLoginAttempts++;
42            System.out.println("Login failed.");
43        }
44    }
45
46    public void lockCheck() {
47        if (failedLoginAttempts >= 3) {
48            accountStatus = "Locked";
49            System.out.println("Warning: Account is locked.");
50        } else {
51            System.out.println("Account is active.");
52        }
53    }
54
55    public static void main(String[] args) {
56        UserAccount user1 = new UserAccount(
57            "Navitha",
58            "navitha@gmail.com",
59            "Active",
60            0,
61            "abc123"
62        );
63
64        System.out.println("Username: " + user1.getUsername());
65
66        user1.setFailedLoginAttempts(-2);
67
68        user1.login("abc123");
69        user1.lockCheck();
70
71        user1.login("wrong1");
72        user1.login("wrong2");
73        user1.login("wrong3");
74
75        user1.lockCheck();
76    }
77 }
```

```
ng\Code\User\workspaceStorage\7e18229a918905c06ac185b
Username: Navitha
Invalid value: Failed login attempts cannot be negati
Login successful.
Account is active.
Login failed.
Login failed.
Login failed.
Warning: Account is locked.
PS D:\NSBM\1 Year\2 Sem\Java\In class\1> █
```

Q3

```

1  public class ThreatLog {
2      int score1;
3      int score2;
4      int score3;
5      int score4;
6      int score5;
7
8      ThreatLog(int s1, int s2, int s3, int s4, int s5) {
9          score1 = s1;
10         score2 = s2;
11         score3 = s3;
12         score4 = s4;
13         score5 = s5;
14     }
15
16     String getClassification(int score) {
17         if (score >= 80) {
18             return "Critical";
19         } else if (score >= 60) {
20             return "High";
21         } else if (score >= 40) {
22             return "Medium";
23         } else {
24             return "Low";
25         }
26     }
27
28     void printReport() {
29         System.out.println("Score 1: " + score1 + " - " + getClassification(score1));
30         System.out.println("Score 2: " + score2 + " - " + getClassification(score2));
31         System.out.println("Score 3: " + score3 + " - " + getClassification(score3));
32         System.out.println("Score 4: " + score4 + " - " + getClassification(score4));
33         System.out.println("Score 5: " + score5 + " - " + getClassification(score5));
34     }
35
36     int getHighestThreat() {
37         int highest = score1;
38
39         if (score2 > highest) {
40             highest = score2;
41         }
42
43         if (score3 > highest) {
44             highest = score3;
45         }
46
47         if (score4 > highest) {
48             highest = score4;
49         }
50
51         if (score5 > highest) {
52             highest = score5;
53         }
54
55         return highest;
56     }
57
58     int countCritical() {
59         int count = 0;
60
61         if (score1 >= 80) {
62             count++;
63         }
64
65         if (score2 >= 80) {
66             count++;
67         }
68
69         if (score3 >= 80) {
70             count++;
71         }
72
73         if (score4 >= 80) {
74             count++;
75         }
76
77         if (score5 >= 80) {
78             count++;
79         }
80
81         return count;
82     }
83
84
85
86     public static void main(String[] args) {
87         ThreatLog log1 = new ThreatLog(91, 55, 38, 74, 62);
88
89         log1.printReport();
90
91         System.out.println("Highest Threat Score: " + log1.getHighestThreat());
92         System.out.println("Critical Threat Count: " + log1.countCritical());
93     }
94
95 }

```

```
ng\Code\User\workspaceStorage\7e182
Score 1: 91 - Critical
Score 2: 55 - Medium
Score 3: 38 - Low
Score 4: 74 - High
Score 5: 62 - High
Highest Threat Score: 91
Critical Threat Count: 1
PS D:\NSBM\1 Year\2 Sem\Java\In cla
```

Q4

```
1  class SecurityAnalyst {
2      static int totalAnalysts = 0;
3
4      String analystID;
5      String name;
6      String specialization;
7      int clearanceLevel;
8
9      SecurityAnalyst() {
10         analystID = "Unknown";
11         name = "Unknown";
12         specialization = "General";
13         clearanceLevel = 1;
14         totalAnalysts++;
15     }
16
17     SecurityAnalyst(String id, String analystName, String spec, int level) {
18         analystID = id;
19         name = analystName;
20         specialization = spec;
21         clearanceLevel = level;
22         totalAnalysts++;
23     }
24
25     void getProfile() {
26         System.out.println("Analyst ID: " + analystID);
27         System.out.println("Name: " + name);
28         System.out.println("Specialization: " + specialization);
29         System.out.println("Clearance Level: " + clearanceLevel);
30     }
31
32     static int getTotalAnalysts() {
33         return totalAnalysts;
34     }
35 }
36
37 class Main {
38     public static void main(String[] args) {
39         SecurityAnalyst analyst1 = new SecurityAnalyst();
40
41         SecurityAnalyst analyst2 = new SecurityAnalyst(
42             "A001",
43             "John",
44             "Malware Analysis",
45             4
46         );
47
48         analyst1.getProfile();
49
50         System.out.println();
51
52         analyst2.getProfile();
53
54         System.out.println();
55
56         System.out.println("Total Analysts: " + SecurityAnalyst.getTotalAnalysts());
57     }
58 }
```

ng\code\user\workspace\stage\7c10225d510505c00dc10.

Analyst ID: Unknown

Name: Unknown

Specialization: General

Clearance Level: 1

Analyst ID: A001

Name: John

Specialization: Malware Analysis

Clearance Level: 4

Total Analysts: 2

○ PS D:\NSBM\1 Year\2 Sem\Java\In class\1>